Agentic AI for IP Network Operations

A Network Management Vendor Perspective
why Agentic AI Demands New Protocol Foundations

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Why Does IP Network Management Need Agents?

Limitations of Traditional Methods

Human Triage is the Bottleneck

Event velocity outpaces human reaction; millions of daily logs and alarms are overwhelming.

Lack of Cross-Domain Correlation

IP, transport, and access domains operate in silos, preventing end-to-end fault diagnosis.

Scripts Break at Scale

Runbooks rely on known failure modes and fail with novel fault combinations.

Unique Value Enabled by Agents

Goal-directed, Multi-step Reasoning

Decomposes goals like 'restore service' into diagnosis, scoping, remediation, and verification steps, adapting dynamically to the observed state.

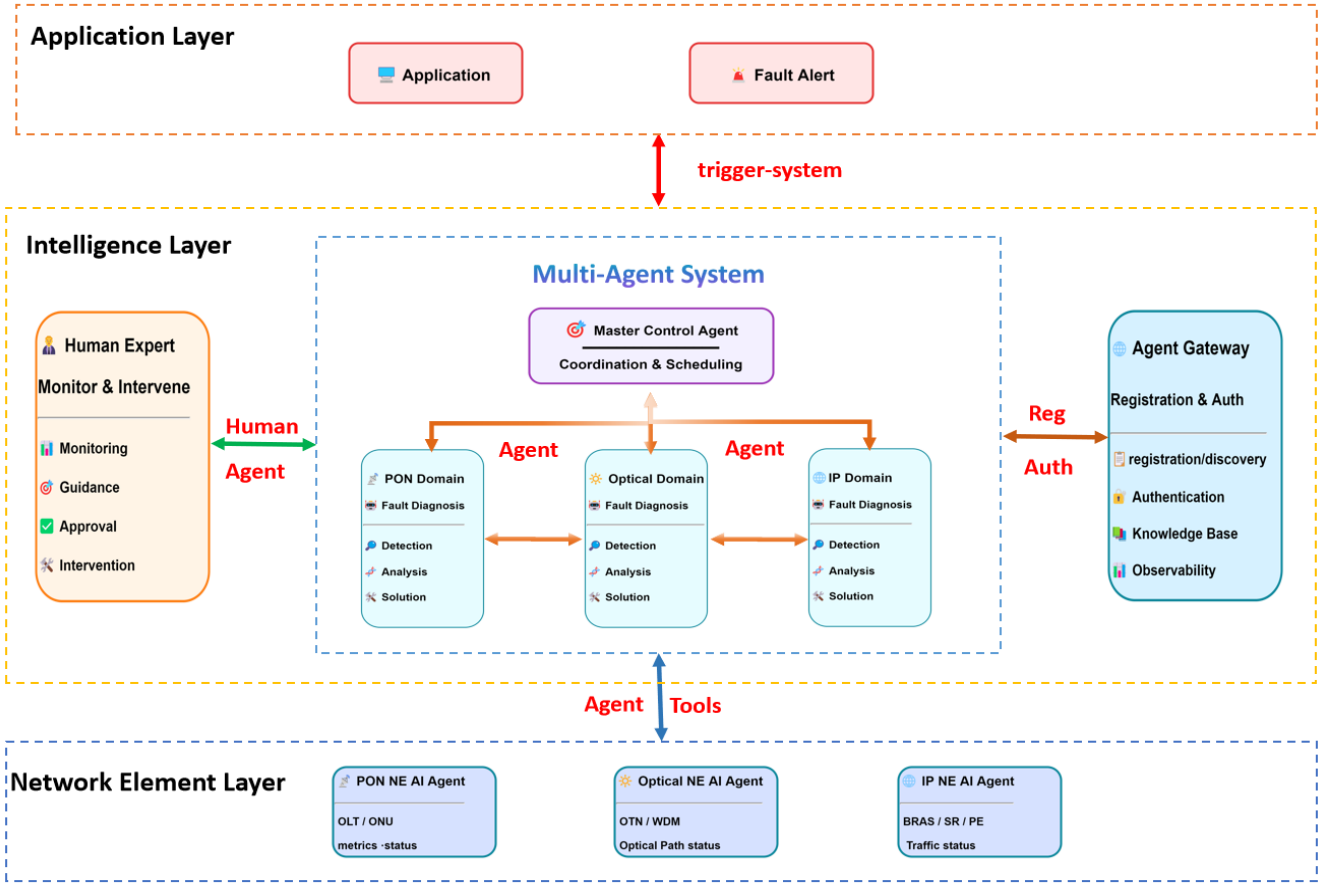
Cross-System Context Retention

Agents maintain task context across domains, enabling correlation analysis and end-to-end diagnosis of cross-domain faults.

Continuous, Proactive Operation

Agents monitor, predict, and act autonomously, shifting from reactive to anticipatory operations.

Multi-Agents：Cross-Domain End-to-End Fault Diagnosis



Workflow: End-to-End Diagnosis Process

01. Fault Trigger

A complaint or alarm triggers a diagnosis intent, sent to the Master Control Agent.

02. Task Splitting

The Master Agent decomposes the task into subtasks and distributes them to PON/Optical/IP Domain Agents.

03. Collaboration

Intra-domain agents collaborate for detection, analysis, and resolution; inter-domain agents synchronize fault data for fusion.

04. Root Cause Identification

The Master Agent aggregates results to determine the end-to-end root cause.

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Inter-Agent Collaboration Interfaces

Current Gaps

- Proprietary protocols block cross-vendor/cross-domain collaboration.
- No unified standard for task lifecycle & data exchange.
- Missing standardized capability description & discovery mechanism

Standardization Focus

- Agent capability description model.
- Task negotiation & state synchronization protocol.
- Unified semantic data exchange format.
- Cross-domain secure context transfe

Goal

🎯Plug-and-play cross-vendor multi-agent collaboration.

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Agent-Tool Collaboration Interfaces

Current Gaps

- Heterogeneous tool APIs require heavy custom adaptation.
- No dynamic capability discovery & invocation standard.

Standardization Focus

- Unified tool capability description & discovery.
- Standard sync/async invocation protocol.
- Task context transfer & dynamic tool registration

Goal

🎯 Low-complexity "Tool-as-a-Service" integration.

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Human-Agent Collaboration Interfaces

Current Gaps

- Fragmented interaction modes with no unified model.
- No standard for intent delivery, approval, intervention & explainability

Standardization Focus

- Structured intent expression model.
- Standard approval & intervention interfaces.
- Explainable output format & human feedback mechanism

Goal

Trusted, controllable human-agent collaboration framework.

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What We Are Asking the IETF to Consider

Standardized Agent Collaboration Protocols

Define protocols for capability description, task negotiation, and secure data exchange between autonomous agents.

Agent-Tool Interaction Interfaces

Establish plug-and-play interfaces to ensure seamless integration of diverse network tools and utilities.

Human-Agent Collaboration Models

Standardize interaction models to ensure effective human control, oversight, and trust in AI-driven operations.

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Next Steps

- Further reviews and comments are welcome and appreciated.

Thanks!